TalentAI: Multi-Agent Intelligent Recruitment System

Project Synopsis -- Problem, Solution, Target User & Scope

1. Problem Statement

Modern corporate recruitment and candidate screening workflows are severely bottlenecked by large volumes of unstructured applicant data. Conventional Applicant Tracking Systems (ATS) rely primarily on rigid keyword matching, which produces two critical failure modes:

- **High False Positives:** Unqualified candidates often engage in 'keyword stuffing' to artificially game ATS algorithms, inundating recruiters with applicants who lack demonstrable technical competency.
- **High False Negatives:** Exceptionally qualified candidates are discarded simply because they describe equivalent skills using alternate phrasing, modern synonyms, or specialized acronyms (e.g., 'k8s' vs. 'Kubernetes').
- **Cognitive Fatigue & Inconsistent Evaluation:** Human recruiters manually reviewing hundreds of multi-page resumes face extreme fatigue, leading to subjective, non-standardized qualification bars and multi-week turnaround times.

2. Proposed Solution: The TalentAI Platform

TalentAI is an autonomous multi-agent recruitment and candidate ranking platform that transforms technical hiring into a transparent, deterministic, and agentic workflow. Rather than delegating recruitment to a single prompt, TalentAI distributes operations across six specialized, collaborating agents:

- **Orchestrator Agent:** Coordinates execution flow, maintains canonical pipeline state, detects inter-agent conflicts, and triggers adaptive replanning.
- **Job Analysis Agent:** Decomposes unstructured job descriptions into non-negotiable mandatory skills, preferred qualifications, and minimum experience thresholds.
- **Resume Analysis Agent:** Parses unstructured CVs in parallel, validates technical competencies, and indexes dense embeddings into ChromaDB vector space.
- **Matching & Ranking Agent:** Executes semantic RAG retrieval and computes composite match scores (Mandatory 60%, Preferred 30%, Experience 10%).
- **Candidate Evaluation Agent (Dual-Stage Hybrid Critic):** Independently challenges shortlists by coupling deterministic rule checks (100% mandatory compliance) with qualitative LLM auditing.
- **Recruitment Communication Agent:** Crafts bespoke interview invitations highlighting specific achievements for approved candidates, and constructive rejection notices.

3. Target User Persona

TalentAI is engineered for **Corporate Talent Acquisition Teams, Technical Recruiters, Engineering Hiring Managers, and High-Growth SMEs**. These users manage technical job openings with high applicant-to-hire ratios and require rapid, bias-free, and verifiable first-round candidate shortlisting with explainable audit trails.

4. Project Scope & CA2 Evolution Notes

Consistency with Approved CA2 Description: TalentAI fulfills the approved CA2 objective of automating candidate screening and ranking via specialized agents, semantic RAG retrieval, and structured scoring.

Explicit Scope Enhancements during CA3 Realization:
In response to academic review comments and to demonstrate genuine multi-agent autonomy, the scope was deliberately upgraded in three key dimensions:

- **Dual-Stage Hybrid Critic Architecture:** Upgraded the Evaluation Agent from a superficial LLM check to a dual-stage system: Stage 1 deterministically validates that 100% of non-negotiable mandatory criteria are satisfied in Python code; Stage 2 applies Groq LLaMA-3.3-70B for nuanced architectural and leadership review.
- **Dynamic Replanning & Conflict Arbitration:** Integrated an autonomous negotiation loop. When the Evaluation Critic rejects a candidate, the Orchestrator executes strategic replanning -- selectively relaxing preferred requirements while strictly holding mandatory requirements 100% inviolable, with a bounded ceiling of 2 rounds.
- **Real-Time Telemetry Streaming:** Built a full-stack telemetry interface utilizing FastAPI Server-Sent Events (SSE) and Next.js 14, streaming chronological agent reasoning steps directly to an interactive recruiter dashboard.